

Real-Time Analytics & Dashboards

Why Visualization Matters

- AIoT insights must be clear, actionable, and easy to interpret.

What Dashboards Display

- 📊 Live sensor data & operational metrics
- 🤖 AI predictions & anomaly alerts
- ❤️ System health & performance status

Operational Benefits

- 👁️ Monitor performance in real time
- ⚠️ Detect issues & respond quickly
- 📈 Support informed decision-making

Automation & Alerts

- 🔔 Trigger alerts for critical events
- ⚙️ Initiate automated workflows & responses



Dashboards transform AI insights into actionable decisions

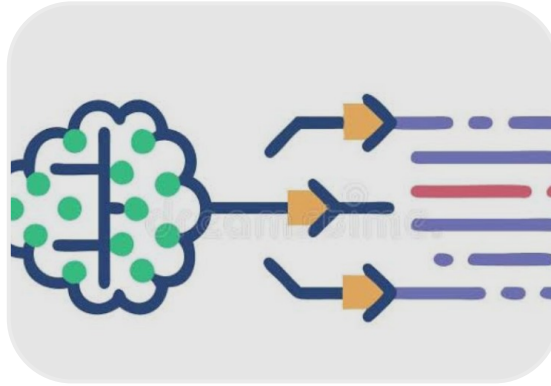
Core Components of AIoT Real-Time Systems

A robust AIoT dashboard system consists of four main layers:



Data Ingestion

- MQTT • Kafka • AWS IoT Core
- Collect & normalize sensor data streams



Stream Processing & AI

- Apache Flink • Spark Streaming • Databricks
- Real-time analytics & anomaly detection
- Predictive insights in motion



High-Performance Data Storage

- In-memory & time-series databases
- Redis • InfluxDB • TimescaleDB • CrateDB
- Fast read/write for real-time access



Visualization Layer

- Grafana • Power BI • Tableau
- Interactive dashboards & KPI monitoring

Key Features and Capabilities

A robust AIoT dashboard system consists of four main layers:



Predictive Maintenance

- ✂ Forecast equipment failures using AI insights
- ✓ Reduce downtime & maintenance costs



Live Anomaly Detection

- 🚨 Instant alerts for abnormal readings
- ✓ Detect temperature, pressure & vibration risks



Asset & Fleet Monitoring

- 📍 Track performance & location in real time
- 🌐 Map views & operational status monitoring



Customizable Dashboards

- 🧩 Drag-and-drop widgets & role-based views
- ✓ Tailore dashboards for operators, engineers & executives

Top Tools & Platforms for AIoT Dashboards

Choose tools based on scalability, analytics needs, and industry use



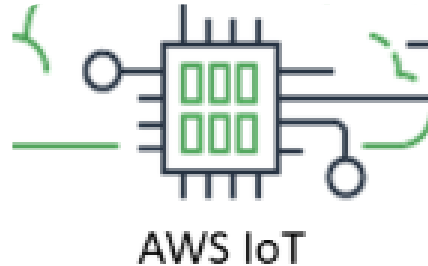
Visualization & Monitoring: Grafana

- Ideal for time-series IoT data visualization
- Integrates with InfluxDB & Prometheus



Unified Analytics & AI Platforms: Databricks

- Lakehouse architecture & real-time analytics
- AI & generative insights



Cloud-Native IoT Dashboards

- **AWS IoT Core + QuickSight**
- **Azure IoT Hub + Power BI**



Industrial IoT Platforms: DATOMS / Kaa

- Industry-specific dashboards & automation



Real-Time IoT Data Query Engine: CrateDB

- High-speed SQL queries for machine & sensor data